VF Asia Footwear Laboratory Testing Manual

Test Method for Water Proof Whole Shoe - Static Test	**Test method # VF- G07** **REF # : TM-R-055** **Page 1 of 4**
Date Issued : March , 2014 **Revision # :**	**Created by : Nila Babia Erazo / Chris Adam**

PURPOSE AND SCOPE:

This method is used to determine the static waterproofness of VF waterproof footwear. The whole shoe static test examines waterproofness of footwear by submerging the footwear in water to a predetermined level for a minimum of four (4) hours. Areas of the footwear tested using this method include: sole to upper bond/seal; upper seams using latex seam seal processes; bootie seams using waterproof membrane construction; as well as the upper fit material(s).

REFERENCE: Timberland FW-R-033, "Whole Shoe Waterproof Requirements".

EQUIPMENT AND MATERIAL:

1. Immersion tank which is large enough to contain the test shoe such that no portion of the test shoe contacts the sides of the tank.
2. Dry weights (rocks , steel)
3. Whole Shoe
4. Steel scale capable of measuring to +/-0.1cm.
5. Water at normal temperature
6. Dry footwear
7. Dry weights (rocks, sandbags, last, etc.)
8. Large rubber bands
9. Logbook
10. Ball-point pen
11. Silver ink pen
12. Pocket knife (or equivalent)
13. Rubbermaid plastic containers (1) 15"W x21"L x 6"D and (1) 15"W x 21"L x 9"D

SAMPLE PREPARATION:

Condition the shoe sample for at least 48 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity to allow the cement to crosslink completely.

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Procedure:

Testing shall be conducted in a standard laboratory environment of 23 ± 2°C and 55 ± 10 % RH.

1. .Remove all hang tags, laces, inserts, and tissue from dry footwear.
2. Fill out logbook with date, type of test, style#, description, factory code and time. If factory code is missing, enter factory name and country of origin, or requester's name.
3. Insert enough weight inside footwear to hold footwear firmly on bottom of the container.
4. Insert footwear to be tested into plastic container.
   - 4.1 Do not add water until all footwear is inside.
   - 4.2 Only test like style# and size in one plastic container.
5. For adult footwear, fill the container with water such that the waterline will be set at 60% of the height of the lowest point on the top line as measured from the bottom of the outsole (Figure 1). For kids footwear, fill the container with water such that the waterline will be set at 60% of the height of the lowest point on the top line as measured from the bottom of the outsole (Figure 2). For both adults and kids footwear, if the lowest point on the top line is greater than 195mm, the water height will be set at 120mm (Figure 2).

Note: The shoe should be constructed such that it is waterproof to a point at least 19mm above the water level line. This point is known as the "High Water Mark" of the shoe.

6 Start test and note starting time in logbook.

7 Manually monitor footwear every 30 minutes, and remove footwear to inspect.

- 7.1 When removing footwear use caution not to drip water into footwear or onto weights. If water contaminates footwear or weights, discontinue the test, dry footwear and/or weights, and retest.
- 7.2 Check for any leakage area greater than 300mm².
- 7.3 If no leaks are detected, record time and elapsed time in the logbook. If test is completed, dry footwear on dryer machine, otherwise proceed to step 3.8 to continue testing.
- 7.4 If leak(s) are detected, record time and elapsed time in logbook. Note area(s) where water penetrated or wicked. Mark wet area(s) with a silver ink pen and dry footwear on dryer machine.
- 7.5 Once dry, as required, perform destructive testing or dismantling of footwear to determine cause of leak.

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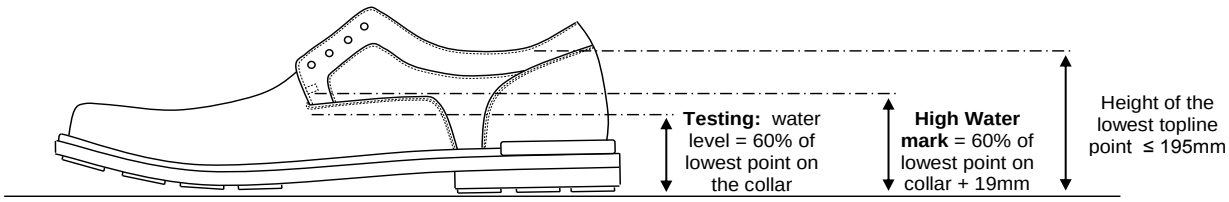
8 .Replace footwear into container(s).

8.1 If water contaminates footwear, discontinue the test, dry footwear, and retest.

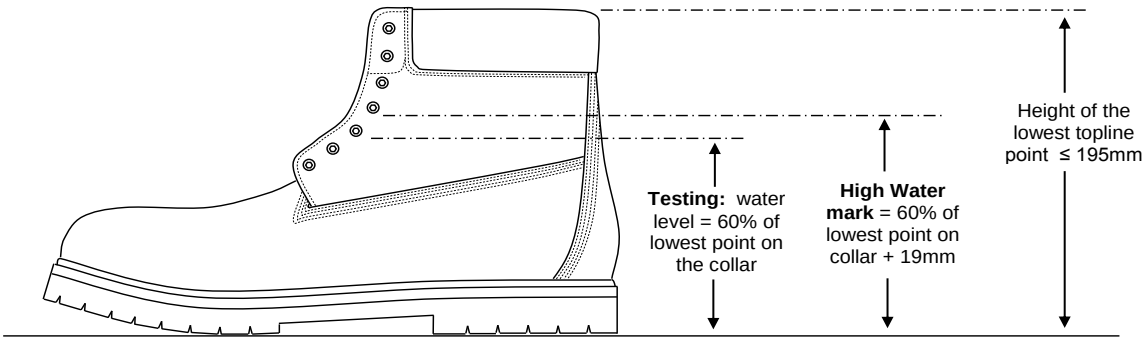
9. After reaching the predetermined number of hours (see Ref 1.1), stop test and remove footwear to inspect according to section # 7.

Reporting

Report test date, SKU#, shoe description, source, season, product stage, a pass/fail rating, results of destructive testing when performed, and any deviations from this procedure.



(a) Oxford



(b) Boot

Figure 1. Location of testing line and high water mark for **adult** shoes with the lowest point of the topline less than 195mm.

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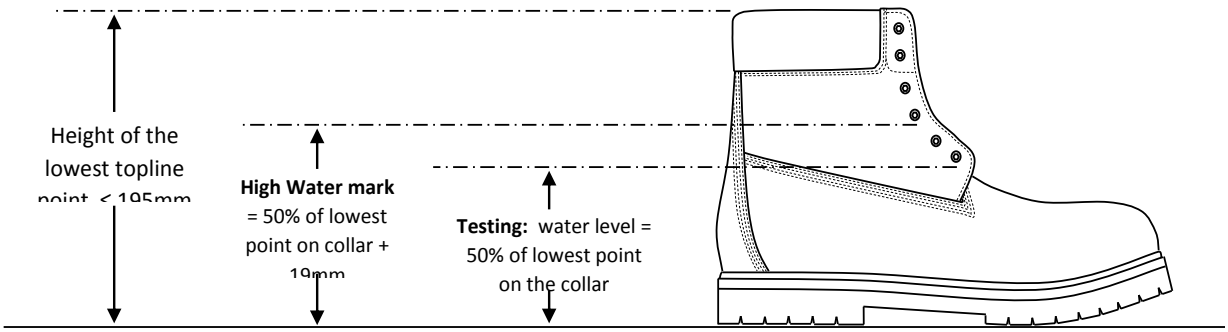


Figure 2. Location of testing line and high water mark for **kids** shoes with the lowest point of the topline less than 195mm.

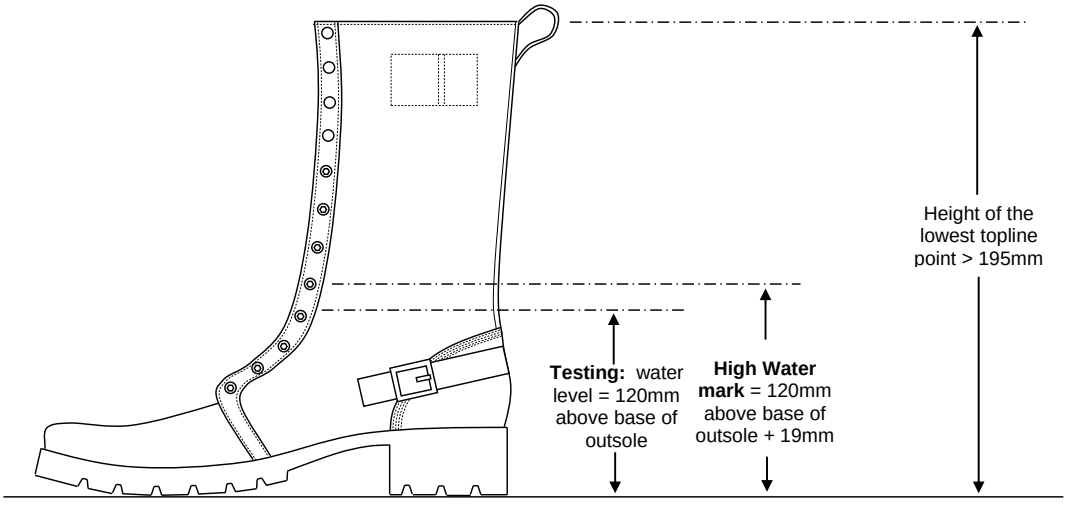


Figure 3 . Location of testing line and high water mark for **adult** and **kids** shoes with the lowest point of the topline greater than 195mm.